

ConvexOptimization 1

王胤雅 Yinya Wang

25114020018

yinyawang25@m.fudan.edu.cn

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PROBLEM I A set C is midpoint convex if whenever 2 points a, b are in C , the average or midpoint $\frac{a+b}{2} \in C$. Obviously a convex set is midpoint convex. It can be proved that under mild conditions midpoint convexity implies convexity. As a simple case, prove that if C is closed and midpoint convex, then C is convex. **PROBLEM II** $a, b \in \mathbb{R}^n$, $a \neq b$. Show $A := \{x \mid \|x - a\|_2 \leq \|x - b\|_2\}$ is a halfspace. Describe it explicitly as an inequality of the form $c^T x \leq d$. Draw a picture.

PROBLEM III Which of the following sets S are polyhedra? If possible, express S in the form $S = \{x \mid Ax \preceq b, Fx = g\}$.

1. $S = \{y_1 a_1 + y_2 a_2 \mid -1 \leq y_1 \leq 1, -1 \leq y_2 \leq 1\}$, where $a_1, a_2 \in \mathbb{R}^n$.
2. $S = \{x \in \mathbb{R}^n \mid x \succeq 0, 1^T x = 1, \sum_{i=1}^n x_i a_i = b_1, \sum_{i=1}^n x_i a_i^2 = b_2\}$, where $a_i, b_j \in \mathbb{R}, 1 \leq i \leq n, j = 1, 2$.
3. $S = \{x \in \mathbb{R}^n \mid x \succeq 0, x^T y \leq 1, \forall y : \|y\|_2 = 1\}$.
4. $S = \{x \in \mathbb{R}^n \mid x \succeq 0, x^T y \leq 1 \forall y : \sum_{i=1}^n |y_i| = 1\}$.

PROBLEM IV Finish the proof of the separating hyperplane theorem: Show that a separating hyperplane exists for C, D , where C, D are convex and $C \cap D = \emptyset$. **PROBLEM V** Let K^* be the dual cone of a convex cone K . Prove the following.

1. K^* is a convex cone.
2. $K_1 \subset K_2$ implies $K_2^* \subset K_1^*$.
3. K^* is closed.
4. $\text{int} K^* = \{y \mid y^T x > 0, \forall x \in K\}$.
5. If $\text{int} K \neq \emptyset$, then K^* is pointed.
6. K^{**} is the closure of K , i.e. $K^{**} = \bar{K}$.
7. If the closure of K is pointed then K^* has nonempty interior.

PROBLEM VI Define monotone nonnegative cone as

$$K_{m+} = \{x \in \mathbb{R}^n \mid x_1 \geq x_2 \geq \cdots \geq x_n \geq 0\}.$$

1. Show K_{m+} is a proper cone.
2. Find K_{m+}^* .